

Netflix Recommendation System

Personalized Content Discovery using Collaborative Filtering

Cult Open Projects 2026 | AI/ML Track

DATASET

Netflix Prize Dataset

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Exploratory Data Analysis

Key insights from the Netflix Prize Dataset

24.8M

Total Ratings

175K

Active Users

9,210

Movies

98.47%

Sparsity

Rating Distribution

Rating	Count	Share
1	1,144,746	4.6%
2	2,537,159	10.2%
3	7,201,212	29.0%
4	8,351,564	33.6%
5	5,584,097	22.5%

Critical Insight: At 98.47% sparsity, only 1.5% of user-movie pairs have ratings — traditional similarity methods fail. This motivates Latent Factor Models.

Approach & Methodology

Two-model comparison strategy

Model 1 — SVD

- Factorises ratings into user & item latent vectors
- Optimised via Stochastic Gradient Descent
- Gold standard for explicit rating prediction
- 200 latent dimensions, 50 epochs

Model 2 — ALS

- Alternates solving for user and item factors
- Fully parallelisable across CPU/GPU cores
- Production standard at Netflix and Spotify
- 200 latent dimensions, 40 iterations

Pipeline: Raw Data → Parser → Active User Filter → 80/20 Split → SVD + ALS → Evaluation

Evaluation Metrics

Measuring prediction accuracy and ranking quality

RMSE

Root Mean Square Error

Penalises large prediction errors quadratically. Lower is better.

MAP@10

Mean Avg. Precision @ 10

Measures ranking quality of the Top-10 list. Relevance: rating ≥ 3.5 ★

Precision@10

% of Top-10 that are relevant

How many recommended movies did the user actually like?

Recall@10

% of liked items captured

Of all movies the user liked, how many did we recommend?

NDCG@10

Position-weighted ranking

Rewards placing the most relevant items at the top of the list.

Experimental Results

SVD vs ALS performance comparison

Metric	SVD	ALS	Winner
RMSE	0.8691	2.806	SVD ✓
MAP@10	0.8537	0.7126	SVD ✓
Precision@10	0.6471	0.5371	SVD ✓
Recall@10	0.6816	0.6126	SVD ✓
NDCG@10	0.9266	0.8515	SVD ✓

0.869

RMSE (SVD)

85.4%

MAP@10 (SVD)

92.7%

NDCG@10 (SVD)

79.8%

Accuracy

- SVD RMSE of 0.8691 matches Netflix Prize competition-grade accuracy
- ALS trains faster and is preferred in production for scalability despite lower offline accuracy

Recommendation Examples

Real outputs from the SVD model

✓ Success Cases

User	Movie	Predicted	Actual
1473980	Pirates of the Caribbean	4.58	5.0 ★
1473980	Freaky Friday	4.81	5.0 ★
716173	20K Leagues Under the Sea	5.00	5.0 ★

✗ Failure Cases

User	Movie	Predicted	Actual
305344	Music of the Heart	4.14	1.0 ★
752642	Lord of the Rings	4.15	2.0 ★

Explainable AI

"We recommend 'Man on Fire' because you enjoyed Lethal Weapon (5/5), Minority Report (5/5), Paycheck (5/5). Users with similar taste also highly rated this movie."

Key Insights & Engineering Learnings

Technical, business, and production perspectives

Technical

- 98.47% sparsity makes k-NN approaches impractical
- Vectorised NumPy ops: 1000× speedup vs Python loops
- SVD biases separate systematic effects from true taste
- SVD predicts ratings; ALS predicts engagement

Business

- Long-tail surfacing is where real business value lives
- Explainability increases user trust and click-through
- ALS preferred in production for scalability at scale

ML Engineering

- Pre-compute & cache for sub-100ms serving latency
- A/B test SVD vs ALS on real engagement metrics
- Popularity-based recs for cold-start users (<20 ratings)

Future Work & What's Next

Roadmap for improving the recommendation system

1. Neural Collaborative Filtering

PyTorch-based NCF to learn non-linear user-item interactions beyond dot products

2. Hybrid Models

Combine content features (genre, cast) with collaborative filtering to solve cold-start

3. Temporal Dynamics

SVD++ with time decay to capture evolving user preferences over months and years

4. Full Dataset Training

Train on all 100M+ ratings with high-RAM infrastructure for maximum accuracy

5. Implicit Feedback

Integrate watch time, browsing history, and abandoned playback as negative signals

6. Online Learning

Incremental model updates as new ratings arrive — no costly full retraining cycles